How an electrode's status is decided

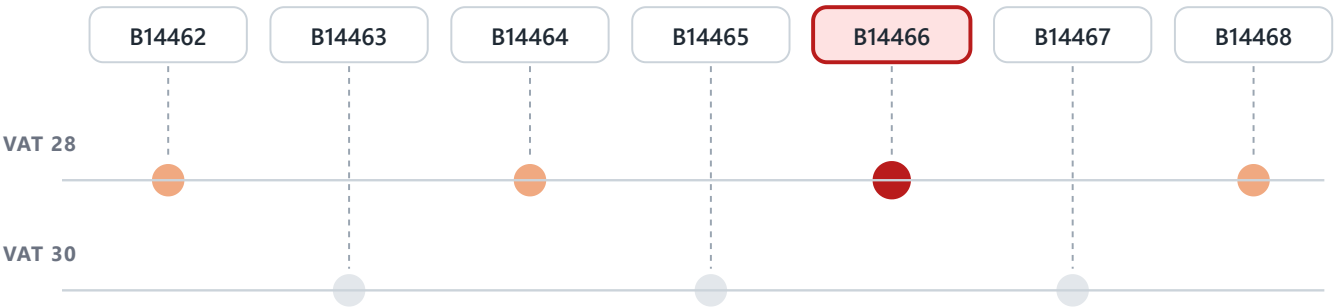
Testing one electrode says something about its neighbours. QC Data Manager, the complete set of propagation rules.

Tested, passed | Tested, failed | Rejected | Scrapped | Needs testing | Not Tested | Test ongoing

1. Everything travels along the vat, not along the line. Electrodes are plated in vats (28, 29, 30) and the line alternates between them, so B14466 and B14464 are neighbours *in vat 28* while B14465 sits between them on the line in a different vat. A fault is the vat's doing, so it spreads to the electrodes that vat made, and skips over everything else.

2. A measurement always beats a guess. Only two statuses are measurements: Tested passed and Tested failed. Somebody put those electrodes on a rig. Everything else is inferred from them, and an inference can always be replaced by a better one.

THE PRODUCTION LINE, IN ORDER



B14466 fails. The doubt travels along the vat 28 lane only, so B14464 and B14468 are scrapped while B14465 and B14467, its neighbours on the line, are left alone.

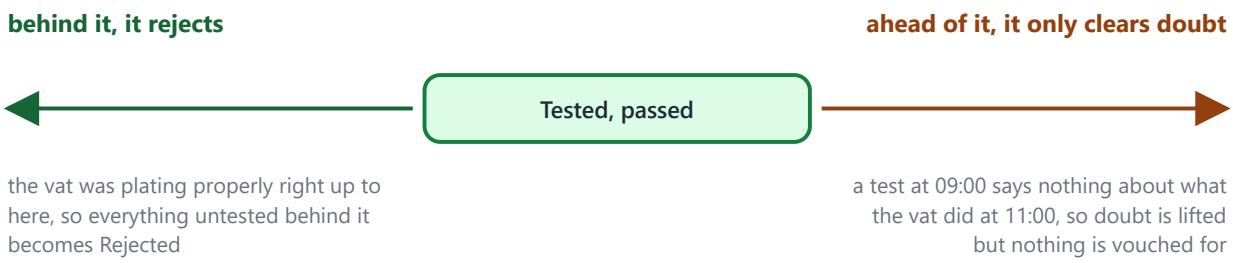
The six statuses

STATUS	MEANING	MEASURED?
Tested, passed	put on a rig, it was good	yes
Tested, failed	put on a rig, it was bad	yes
Rejected	presumed good, because of a pass nearby	no, a guess
Scrapped	presumed bad, because of a fault nearby	no, a guess
Needs testing	in doubt, somebody should test it	no, a request
Not Tested	nothing is known about it	no

Plus one that is never stored: **Test ongoing**, which means the electrode is on the QC board right now. It is worked out fresh every time the screen is drawn.

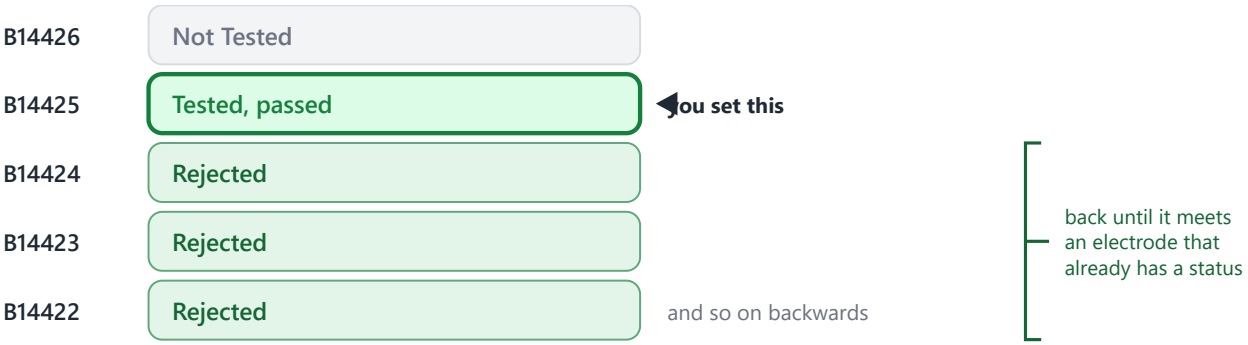
Scenario 1: you set an electrode to Tested, passed

A pass does two different things in the two directions, and they are not symmetric. **Behind it, it rejects. Ahead of it, it only clears doubt.**



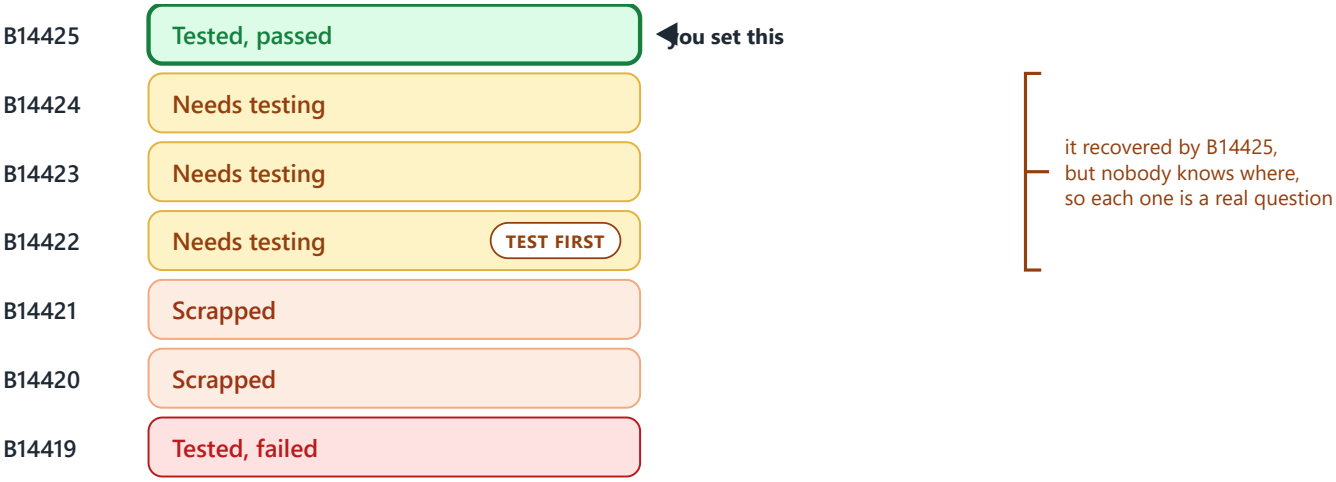
1a. Behind it, nothing in the way

Everything untested behind it in the same vat becomes Rejected, back until it meets an electrode that already has a status.



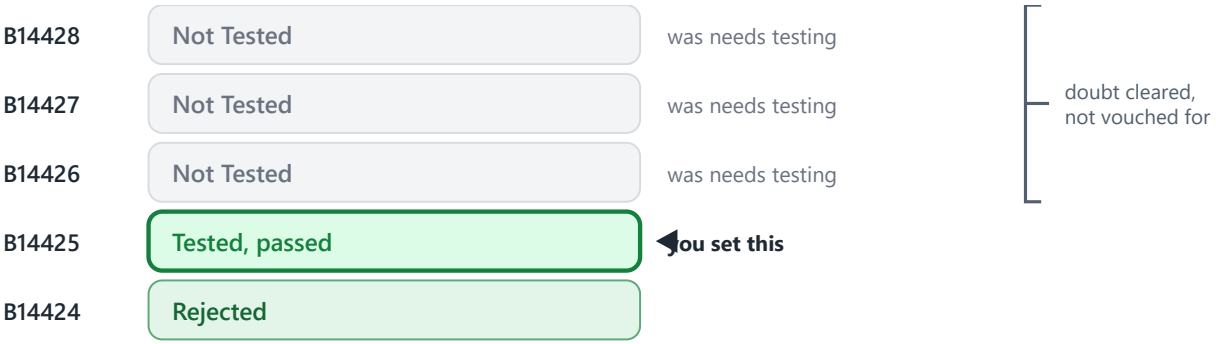
1b. Behind it, with a fault in the way

Nothing is rejected. The stretch between the fault and the test stays in doubt and is flagged for testing instead. We now know production had recovered *by* B14425, but not *where* it recovered, so every electrode in between could be on either side of that line.



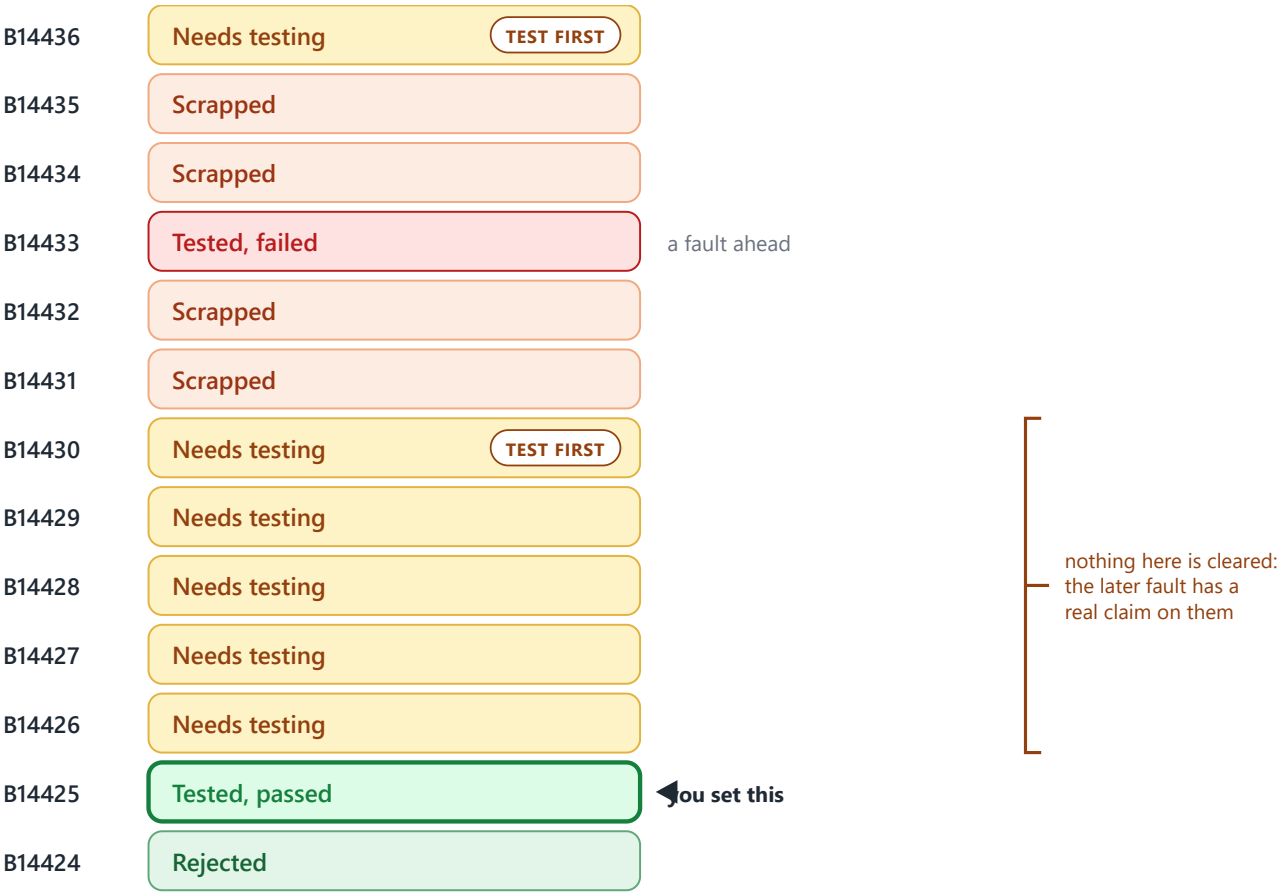
1c. Ahead of it, no fault ahead

Electrodes ahead that were flagged Needs testing go back to Not Tested. They were doubted because of a fault the pass now stands between, so that doubt can no longer reach them. They do **not** become Rejected: they were plated after the test was done, so nothing here vouches for them.



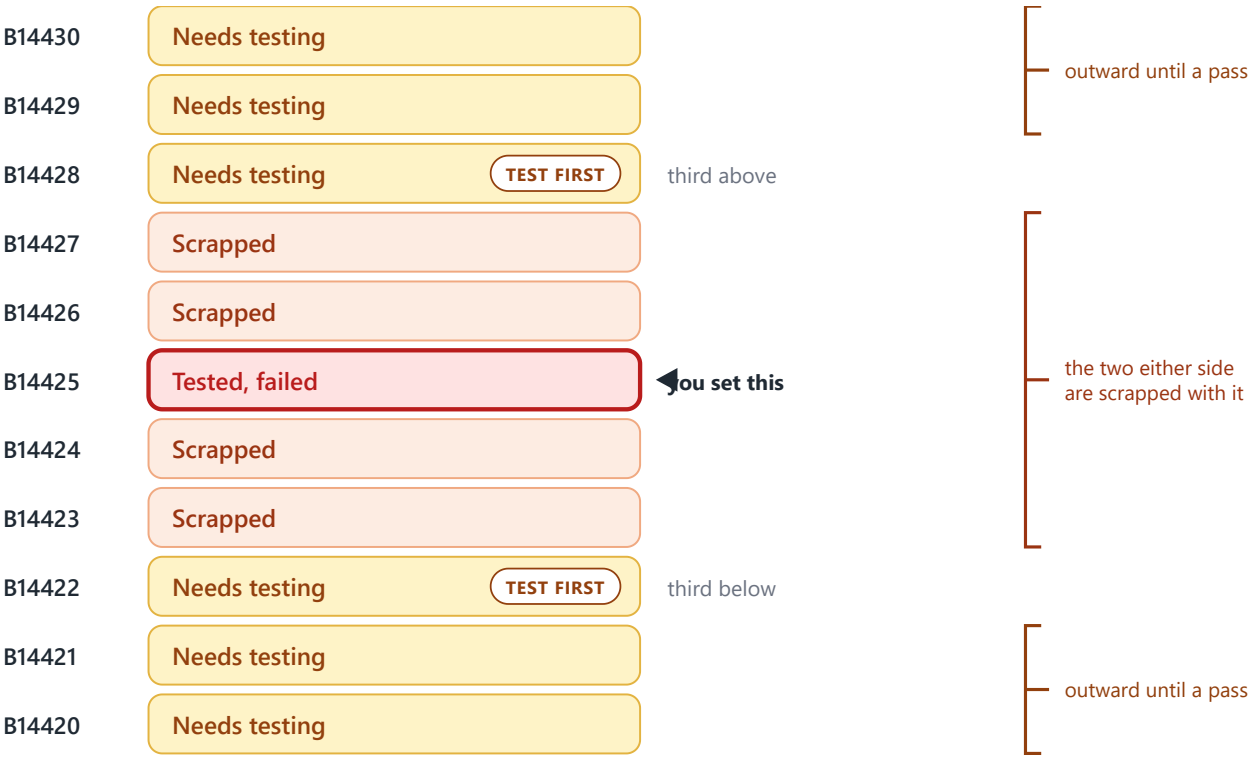
1d. Ahead of it, with a fault ahead

Nothing ahead is cleared. Those electrodes are in doubt because of the *later* fault, and that doubt is real.



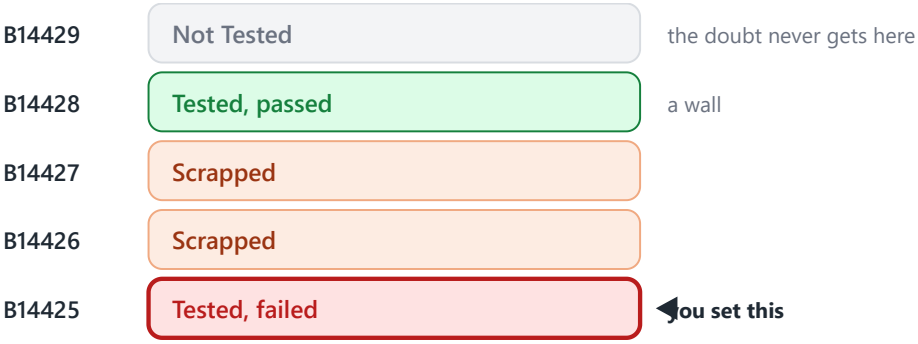
Scenario 2: you set an electrode to Tested, failed

A fault spreads **both ways equally**, along its own vat. The two either side are scrapped with it. The third each way is flagged Needs testing and marked **test first**, because those are where the fault most likely started and stopped, so they are the ones worth testing next. Beyond that it keeps marking Needs testing outward until it reaches an electrode that was tested and passed.



2a. A pass stops it

The doubt cannot travel through an electrode somebody tested and passed.



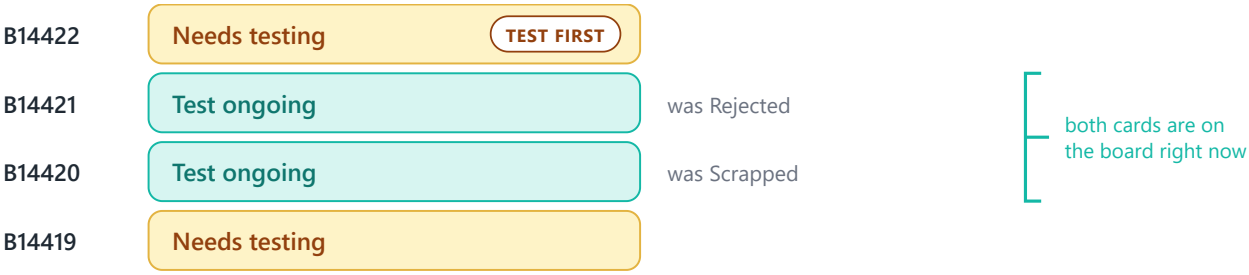
2b. Other vats are untouched

The line alternates between vats. Only vat 28 is affected here. The same is true of a pass: it rejects along its own vat and leaves the others alone.



Scenario 3: you put an electrode on the QC board

It reads **Test ongoing**, whatever it read before. A recorded Scrapped, a Rejected, a Needs testing, all of them give way: putting it on the board is the statement that its status is being revisited. It also loses any **test first** flag, because somebody is already testing it.



3a. Taking it back off the board

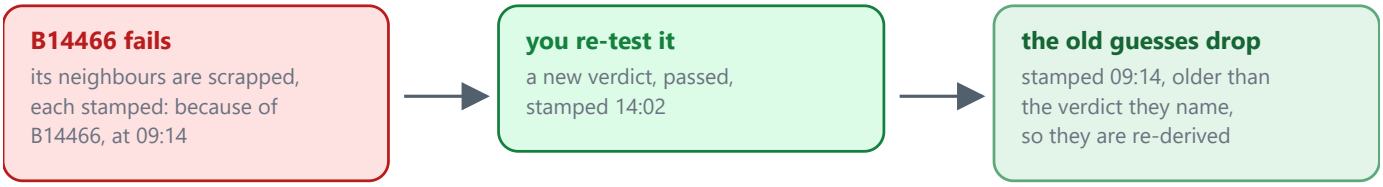
Test ongoing is never saved anywhere. Remove the card and the electrode reads whatever it *would* have read if it had never been on the board. Note this is not always the status it showed before. B14420 had a verdict of its own, so it comes back. B14421 only had a guess, and by the time the card came off, a nearby fault had a better claim on it.

B14421	Needs testing	reopened by the fault above
B14420	Scrapped	back to its own verdict

Scenario 4: you re-test something

Setting a new verdict on an electrode that already had one throws away every guess the old verdict produced, and re-derives them from the new one. So if B14466 failed, condemning its neighbours, and you later re-test it and it passes, those neighbours are not left scrapped. They are worked out again.

This is why the app records, for every guessed status, **which** verdict it came from and **when**. A guess older than the verdict it names was drawn from a version of that verdict that no longer exists, so it is discarded.



The rules underneath

What a sweep may overwrite

STATUS	CAN A SWEEP CHANGE IT?	BECAUSE
Not Tested	yes	nothing is known, so anything is an improvement
Needs testing	yes	it is a request, not an answer
Rejected	yes	nobody measured it
Scrapped	no, it keeps it, but the sweep carries on past	a recorded verdict
Tested, failed	no, and the sweep carries on past	a measurement
Tested, passed	no, and it stops the sweep dead	the only wall

A Rejected electrode being overwritten is the surprising one. It matters because a Rejected electrode never appears in anybody's work queue. If a fault turned up beside it and it kept that status, it would ship without ever being looked at.

Only a real pass is a wall

A guess does not bound another guess. Rejected does not stop doubt, only Tested passed does. A scrapped electrode does not stop a sweep either, so two nearby faults produce one continuous zone rather than two separate ones.

Distances

RULE	REACH
scrapped either side of a fault	**2** electrodes
test first flag	the **3rd** each way
how far Needs testing runs	until a Tested passed

If the process log does not say which vat an electrode came from, the fault falls back to counting along the production line instead, and stops after the third. A guess about the line is worth two electrodes, not a whole stretch.

Why a row is locked

A row is locked when a card **currently on the QC board** is what decides its status. Dragging that card somewhere else would change the row, so it cannot be pinned by hand at the same time.

Two things that look similar and are not. **Because of** names **one** decision, the one recorded against the status. **Locked by** covers **every** decision that contributed. A status can genuinely have more than one cause: a stretch cleared between two passes needs both of them to stay cleared, so both lock it. A finalised verdict never locks anything, it cannot be dragged anywhere.

Generated from docs/status-propagation.md by tools/generate_status_propagation_pdf.py. Written with the planned wording Rejected in place of the current Released.